

Publication List Leander Thiele

August 31, 2026

- Y. Itow, J. Liu, H. Maesaka, V. Mikuni, N.-M. Nguyen, H. Miyataka, A.J. Nishizawa, P. de Perio, D. Ratner, K. Terao, **L. Thiele**, O. Alterkait, F. Drielsma, R. Garcia, M. Iwasaki, A. Hafizhu Shali, F. Tarsitano, T. Terada, J. Xia, *Future of Artificial Intelligence for Science in Japan 2024 Community Report*, 2026, [arXiv:2608.27807 \[hep-ph\]](#)
- J. Liu, V. Krishnaraj, K. Vovk, K. Aizawa, A.E. Bayer, L. Blot, J. Cowell, S. Garg, J. Gree, A. Hell, B. Horowitz, M. Ichikawa, K. Iemoto, K. Kondo, Z. Lorsin, K. McCarthy, J. Robinson, M. Ruiz-Granda, **L. Thiele**, I. Vovk, M. Zhou, *AI's Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology II: Project Planning and Proposal Evaluation*, 2026, [arXiv:2607.25881 \[cs.CL\]](#)
- A. Hell, K. Vovk, V. Krishnaraj, J. Liu, K. Aizawa, A.E. Bayer, L. Blot, J. Cowell, S. Garg, J. Gree, B. Horowitz, M. Ichikawa, K. Iemoto, K. Kondo, Z. Lorsin, K. McCarthy, J. Robinson, M. Ruiz-Granda, **L. Thiele**, I. Vovk, M. Zhou, *AI's Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology I: Literature Review*, 2026, [arXiv:2607.25672 \[astro-ph.IM\]](#)
- B.Y. Wang, **L. Thiele**, M. Ho, *Cluster Mass Inference from Galaxy Kinematics*, 2026, [arXiv:2606.12938 \[astro-ph.CO\]](#)
- L. Thiele**, *Machine Learning Techniques for Astrophysics and Cosmology: Simulation-Based Inference*, 2026, [arXiv:2605.10719 \[astro-ph.CO\]](#), Invited chapter for the edited book "Machine Learning Techniques for Astrophysics and Cosmology" (Eds. C. Bambi, V. Kashyap, S. Shashank, N. Yoshida, Springer Singapore, expected in 2026)
- A. Hell, **L. Thiele**, *LLMs with in-context learning for Algorithmic Theoretical Physics*, 2026, [arXiv:2605.08212 \[cs.LG\]](#), poster at ICML 2026 AI4Physics workshop
- L. Thiele**, *Bayesian Cosmic Void Finding with Graph Flows*, 2026, *OJAp* 9, [arXiv:2602.14630 \[astro-ph.CO\]](#), poster at ICML 2026 AI4Physics workshop
- J. Armijo, **L. Thiele**, J. Liu, *Replicating weak-lensing summary-statistic covariances with normalizing flows*, 2026, *PRD* 114, 023567, [arXiv:2601.20669 \[astro-ph.CO\]](#)
- G. Fabbian, F. Bianchini, A. Sabyr, J.C. Hill, C.C. Lovell, **L. Thiele**, D.N. Spergel, *A new constraint on the y -distortion with FIRAS: implications for feedback models in galaxy formation and cosmic shear measurements*, 2025, [arXiv:2512.03038 \[astro-ph.CO\]](#)
- A. Tokiwa, A.E. Bayer, J. Armijo, J. Liu, R. Terasawa, **L. Thiele**, M. Alvarez, L. Blot, M. Takada, *Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects*, 2025, *JCAP* 07, 023 [arXiv:2511.20423 \[astro-ph.CO\]](#)
- B. Barthe-Gold, N.-M. Nguyen, **L. Thiele**, *Reconstructing the local density field with combined convolutional and point cloud architecture*, 2025, [arXiv:2510.08573 \[astro-ph.CO\]](#), poster at ML4PS workshop NeurIPS 2025

B.Y. Wang, **L. Thiele**, *Set-based Implicit Likelihood Inference of Galaxy Cluster Mass*, 2025, [arXiv:2507.20378 \[cs.LG\]](#), spotlight talk at ML4Astro workshop (colocated with ICML 2025)

J.A. Cowell, J. Armijo, **L. Thiele**, G.A. Marques, C.P. Novaes, D. Grandón, S. Cheng, M. Shirasaki, D. Alonso, J. Liu, *First Constraints from Marked Angular Power Spectra with Subaru Hyper Suprime-Cam Survey First-Year Data*, 2025, MNRAS 546, 2, [arXiv:2507.12315 \[astro-ph.CO\]](#)

L. Thiele, A.E. Bayer, N. Takeishi, *Simulation-Efficient Cosmological Inference with Multi-Fidelity SBI*, 2025, [arXiv:2507.00514 \[astro-ph.CO\]](#), poster at ML4Astro workshop (colocated with ICML 2025)

E. Calabrese, J.C. Hill, H.T. Jense, A. LaPosta, et al (incl. **L. Thiele**), *The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models*, 2025, JCAP 11, 063, [arXiv:2503.14454 \[astro-ph.CO\]](#)

C. Jacobus, **L. Thiele**, P. Harrington, J. Liu, Z. Lukic, *Enhancing Cosmological Simulations with Efficient and Interpretable Machine Learning in the Gabor Wavelet Basis*, 2024, poster at Workshop on Machine Learning and the Physical Sciences (NeurIPS 2024)

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D. Grandón, G.A. Marques, **L. Thiele**, S. Cheng, M. Shirasaki, J. Liu, *Impact of baryonic feedback on HSC Y1 weak lensing non-Gaussian statistics*, 2024, PRD 110, 103539, [arXiv:2403.03807 \[astro-ph.CO\]](#)

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